QUD sensitivity as a parameter of presupposition triggers: Evidence from Korean

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1 Introduction: Context-sensitive presuppositions

Traditionally, not-at-issue projective meanings such as presuppositions are considered to be triggered by a certain lexical item. Some examples:

(1) a. Additivity and low likelihood presuppositions triggered by English *even* (Karttunen & Peters, 1979): ‘Even Miguel loves Paul.’
$\rightsquigarrow$ (i) There is someone else other than Miguel who loves Paul, and (ii) it was less expected that Miguel would love Paul.

b. Factivity presupposition triggered by English *know* (Kiparsky & Kiparsky, 1970; Karttunen, 1971): ‘Miguel knows that Paul is married.’
$\rightsquigarrow$ Paul is married.

c. Repetitive and restitutive presuppositions triggered by English *again* (von Stechow, 1995, 1996): ‘Miguel opened the door again’.
$\rightsquigarrow$ (i) Miguel had opened the door before, or (ii) the door had been open before.

More recently, contextual cues such as the Question Under Discussion (QUD) (Roberts, 2012) have featured in pragmatic accounts of projective meanings (Simons et al., 2010, 2016). Take factive predicates as examples.

(2) Hypotheses about what projects and why: (Simons et al., 2010)

a. All and only those implications of (embedded) sentences which are not-at-issue relative to the Question Under Discussion in the context have the potential to project.

b. Operators (modals, negation, etc.) target at-issue content.

c. Context: A nutritionist has been visiting first grade classrooms to talk to the children about healthy eating.
Q: What most surprised you about the first graders?
A: They didn’t know that you can eat raw vegetables

Stepping back, we might naturally expect the triggering mechanism to differ across presupposition types. Beaver et al. (2017) point out this issue.

> “It remains an open research problem to determine which projective implications can be computed along such lines, without, in Stalnaker’s (1974, p. 61) words, “building anything about presuppositions into the meaning of particular words or constructions.” Contrariwise, it remains an open problem to determine exactly which projective implications can be computed reliably only under the assumption that particular words or constructions lexically trigger a particular presupposition.” (Beaver et al., 2017)

... as well as Karttunen (2016).

> “We will never have a unified account of the phenomena on Levinson’s list of presupposition triggers because they are not all of the same species.” (Karttunen, 2016)

Our empirical observations: Two adverbial iteratives in Korean, *tasi* and *tto*, have different presupposition triggering mechanisms: the former is entirely structure-sensitive, while the interpretation of the latter is parameterized relative to the contextually salient QUD.

Our theoretical claims:

- Presuppositions of the same type are not necessarily either structure-sensitive or context-sensitive. Hybrid-type presuppositions occupying a middle ground, namely partially context-sensitive presuppositions, are also possible.
- Context-sensitivity can be a parameter for a finer-grained taxonomy of presupposition triggers.

2 Iteratives

2.1 Iterative presupposition

We start by defining *iterative* as a linguistic expression that presupposes the existence of an earlier time where a given proposition holds. The English adverb *again* is a good example. The presupposition triggered by an iterative is an *iterative presupposition*.

(3) Miguel visited Barcelona *again*. (#He had never visited Barcelona before.)
Presupposes: There is an earlier time where Miguel had visited Barcelona.

Iteratives in certain languages such as German *wieder* ‘again’ and English *again* are known to have two possible readings: repetitive and restitutive.

(4) Sally hammered the metal flat again. (von Stechow, 1995)

a. Repetitive reading: Sally had flattened the metal at some previous time.
b. Restitutive reading: The metal had been flat at some previous time (and Sally had not necessarily hammered the metal flat before).

The standard approach to these two readings is to construct a uniform semantics for *again* with two possible adjunction sites in the structure (Rapp & von Stechow, 1999; von Stechow, 1995, 1996; Beck, 2005).

(5) a. The Logical Form (LF) of ‘Sally hammered the metal flat’: (von Stechow, 1995)
[[the metal] [1 [$_{\underline{VP}}$ Sally [$_{V'}$ t1 [$_{V'}$ hammered [$_{\underline{SC}}$ PRO1 flat]]]]]

b. Repetitive reading: *again* modifies the entire VP

c. Restitutive reading: *again* modifies the small clause

Narrowing down to the repetitive reading, Myers (2024) reports two different types of repetitive readings triggered by the prefix *s-* in Kanien’kéha (also known as Mohawk).

(6) a. Subjectless presupposition with stative verb
Context: Mary loved John but then they broke up. John began dating Helen and ...
Helen s-a-ho-norónhhkw-e’.
Helen REP-FACT-Fsg>Msg-love-PUNC
‘Helen loved him.’ (Lit. ‘Helen [loved him] again.’)

b. Subjectless presupposition with intransitive verb
Context: John died. Then ...
Kó:r s-a-h-rénhe-ie’.
Paul REP-FACT-MsgA-die-PUNC
‘Paul died.’ (Lit. ‘Paul [died] again.’)

(7) Objectless presupposition with transitive verb
Context: Yesterday, Paul went to his favorite restaurant. He didn’t eat anything all day beforehand. At the restaurant, he ate cherries first. Then ...
Kítkit s-a-ha-’wà:ra-k-e’.
chicken REP-FACT-MsgA-meat-eat-PUNC
‘He ate chicken.’ (Lit. ‘He [ate] chicken again.’)

Myers also supports the structural analysis.

(8) The LF of ‘Paul ate chicken’ in Kanien’kéha:
[$_{\underline{VoiceP}}$ [$_{NP}$ Paul] [$_{Voice'}$ ∅ [$_{\underline{FP}}$ [$_{NP}$ chicken] [$_{F'}$ ∅ [$_{\underline{VP}}$ eat]]]]]

(9) a. Sentential repetitive presupposition: *s-* modifies the entire voiceP

b. Subjectless repetitive presupposition: *s-* modifies only the FP

c. Objectless repetitive presupposition: *s-* modifies only the VP

2.2 Korean iteratives

There are two adverbial iteratives in Korean: *tasi* and *tto*[1]. Here are their four semantic properties.

A. The bisyllabic *tasi* is compatible with both the repetitive and restitutive readings, while the monosyllabic *tto* is only compatible with the repetitive reading. The two iteratives have a difference in their repetitive reading, which will be explained shortly in Item D of this section.

(10) Mikeyl-un **tasi/tto** mwun-ul yel-ess-ta.
Miguel-TOP *tasi/tto* door-ACC open-PST-DEC
'Miguel opened the door again$_{\text{tasi/tto}}$.'
(i) Repetitive presupposition (✓*tasi*, ✓*tto*): 'Miguel had opened the door before.'
(ii) Restitutive presupposition (✓*tasi*, ✗*tto*): 'The door had been open before (and Miguel had not necessarily opened it before).'

B. Only the monosyllabic *tto*, but not the bisyllabic *tasi*, triggers the subjectless/objectless presupposition just as the prefix *s-* in Kanien'kéha does.

(11) a. Subjectless presupposition with intransitive verb
Context: John died. Then ...
Phol-i {**✗tasi/✓tto**} cwuk-ess-ta.
Paul-NOM *tasi/tto* die-PST-DEC
'Paul died.' (Lit. 'Paul [died] again.')

b. Objectless presupposition with transitive verb
Context: Yesterday, Paul went to his favorite restaurant. He didn't eat anything all day beforehand. At the restaurant, he ate cherries first. Then ...
chikhin-ul {**✗tasi/✓tto**} mek-ess-ta.
chicken-ACC *tasi/tto* eat-PST-DEC
'He ate fried chicken.' (Lit. 'He [ate] fried chicken again.')

1. As far as we know, there are two more relevant items which we will not discuss in this presentation: *tolo* and *-to*. *tolo* is an adverb that is compatible with the restitutive reading only. *-to* is a postposition that presupposes the existence of a non-identical alternative of the individual that the NP it attaches to denotes.

(i) Mikeyl-un mwun-**to** yel-ess-ta.
Miguel-TOP door-*to* open-PST-DEC
'Miguel opened the door.' (Lit. 'Miguel also opened [the door].)
Presupposes: Miguel opened an alternative of the door, e.g., a window.

Although the postposition *-to* does not fit our definition of iteratives, it aligns with them in triggering an existence presupposition of a certain type of alternative. Lee (2022) also points out that the monosyllabic *tto* and the postposition *-to* share the same etymological origin.

C. The monosyllabic *tto* can have the additive reading like English *also, then*.[2] For this reason, Myers (2024) considers that *tto* is not a pure repetitive.

(12) Mikeyl-un kul-ul ssu-ko ku kul-ul **tto** ilk-ess-ta.
Miguel-TOP text-ACC write-and that text-ACC *tto* read-PST-DEC
'Miguel wrote a text, and also_{tto} read that text.'

Example (13) presents a key data point for our discussion.

(13) Nan-un yenghwa-nun **tto** cheum po-n-ta.
I-TOP movie-ACC *tto* first.time see-PRES-DEC
'Also, I'm watching a movie for the first time.'

D. The repetitive readings of *tasi* and *tto* do not have exactly the same meaning. Unlike *tto*, *tasi* has the *corrective reading*: the effect of the previous action is nullified (Yoon, 2007).

(14) Mikeyl-un yenkwu-lulwihay cikkwan-ul **tasi/tto** swucipha-yss-ta.
Miguel-TOP research-for judgment-ACC *tasi/tto* collect-PST-DEC
'Miguel collected judgments for his research $\text{again}_{\text{tasi/tto}}$,' and ...
(a) (with *tto*) he had collected judgments at a previous time.
(b) (with *tasi*) he had collected judgments at a previous time, and the previously collected judgments had turned out to be useless for the research.

However, the corrective reading is not a presupposition as it fails to project under a conditional or a negation, unlike the repetitive or restitutive reading.

(15) a. *tasi* under a conditional
Manyak mikeyl-i cikkwan-ul **tasi** swucipha-myen ...
if Miguel-NOM judgment-ACC *tasi* collect-if
'If Miguel collects judgments $\text{again}_{\text{tasi}}$ ...'

b. *tasi* under a negation
Mikeyl-un cikkwan-ul **tasi** swucipha-ci anh-ass-ta.
Miguel-TOP judgment-ACC *tasi* collect-NMZ NEG-PST-DEC
'Miguel did not collect judgments $\text{again}_{\text{tasi}}$.'

(16) a. Repetitive reading (i.e., Miguel had previously collected judgments):
Compatible with both (15a) and (15b)

b. Corrective reading (i.e., previously collected judgments turned out to be useless):
Incompatible with both (15a) and (15b)

We thus argue that *tasi* and *tto* are triggers of the presupposition of the same type: the iterative presupposition (i.e., there is a previous time where a given proposition is true of).

2. Ryu (2018) reports an additive-like useage of *tasi*, which he dubs *sequental reading*. We will not discuss it in this presentation. Refer to Ryu (2018) for details.

3 Analysis

Again, Beaver et al.'s (2017) open question concerns to what extent a presupposition-triggering mechanism can be uniform across different types of presuppositions: which presuppositions are triggered lexically and which are QUD-dependent?

The two adverbial iteratives in Korean –*tasi* and *tto*– provide insight into this dichotomy. Our claims:

- QUD-dependency/sensitivity can be a parameter not only for different types of presuppositions, but also for the same type of presuppositions.
- The taxonomy is not entirely dichotomic and there can be hybrid-type triggers occupying a middle ground, i.e., a lexically triggered presupposition that cannot be completed without a QUD.

Empirical support for our claims:

- Both the bisyllabic *tasi* and the monosyllabic *tto* are lexical presupposition triggers. The presuppositions triggered by these two items are sensitive to the structure to some extent.
- The presupposition triggered by the bisyllabic *tasi* is entirely structure-sensitive.
- The presupposition triggered by the monosyllabic *tto* requires a QUD in order to be completed. It is partially context-sensitive.

3.1 *tasi*

We follow previous analyses in arguing that the repetitive and restitutive reading of *tasi* is entirely structure-sensitive (Oh, 2015; Lee, 2017, 2018; Ryu, 2018; Yoon, 2007). Following the structural approaches for iteratives (von Stechow, 1995, 1996; Beck, 2005), **we argue that the repetitive and restitutive readings of *tasi* are available depending on the adjunction site of the adverb.**

(17) Repetitive/restitutive *tasi*:
$[\![tasi_{\text{rep/res}}\ p(e)]\!] = p(e)$, defined only if $\exists e' \prec e.p(e')$.

All possible readings of *tasi* are entirely structure-sensitive.

(18) Input arguments of *tasi* for its possible readings:

a. Repetitive reading: the VP
b. Restitutive reading: the small clause

3.2 *tto*

At first glance, *tto* resembles German *erneut* 'anew', which is compatible with the repetitive reading but not with the restitutive reading (Rapp & von Stechow, 1999).

(19) Maria hat die Tuer {wieder/erneut} geöffnet.
Maria has the door again opened
'Maria has opened the door again.'
(i) Repetitive presupposition (✓*wieder*, ✓*erneut*): 'Maria had opened the door before.'
(ii) Restitutive presupposition (✓*wieder*, ✗*erneut*): 'The door had been open before (and Maria had not necessarily opened it before).'

A mechanism proposed by von Stechow and Rapp to capture the differences between *wieder* and *erneut* is *the Visibility Parameter*. The adverb *erneut*, unlike *wieder*, is [-Visibility] and cannot attach to the AP whose head is phonologically empty due to decomposition.

(20) The Visibility Parameter for decomposition adverbs (D-adverbs):
A D-adverb can/cannot attach to a phrase with a phonetically empty head.

(21) [Maria $[_{VP}$ [$\emptyset$+open$_{Adj}]_V$ $[_{AP}$ t [the door]]]]

Is this analysis applicable to *tto*? We believe not. A striking piece of evidence comes from the additive reading, especially (13), repeated below as (22).

(22) Nan-un yenghwa-nun **tto** cheum po-n-ta.
I-TOP movie-ACC *tto* first.time see-PRES-DEC
'Also, I'm watching a movie for the first time.'

In (22), *tto* does not express simple addition. **It presupposes a previous event satisfying a proposition that addresses the contextually salient QUD. In other words, the presupposition in (22) is triggered lexically, but cannot be completed without a contextually salient QUD** (e.g., 'What kind of things did you do on dates before?')

(23) Context: Two people, Paul and Miguel, are in a movie theater for their first date. Miguel had been on a date with other people before, but it is his first time having a movie theater date.
Paul: When you were dating before, what kind of things did you do on dates?
Miguel: I did a lot of things. I went on picnics, took winery tours, and went skiing. But (22).

(24) Context: It is Miguel's first time ever to be on a date with someone.
Paul: When you were dating before, what kind of things did you do on dates?
Miguel: Well, I've never been on a date before. #(22).

Now the semantics of *tto* can be formalized as follows.

(25) $[\![tto\ p(e)]\!]^C = p(e)$, defined only if $\exists e' \prec e.\exists q.q(e')$ such that q answers the salient QUD in the context C.

4 Discussions

4.1 Iteration as a common denominator

Recall our definition of an iterative: a linguistic expression that presupposes the existence of an earlier time where a given proposition holds. For simplicity, let us illustrate an iterative as a sentential operator and use p to refer to the input proposition and q to refer to the proposition that is true of an earlier event.

(26) $[\![Op_{iter} p(e)]\!] = p(e)$, defined only if $\exists e' \prec e.q(e')$, where q is... what?

Intuitively, the term *a given proposition*, represented as q in (26), cannot remain unrestricted. The standard restriction arises from the structural analysis of the repetitive and restitutive readings: it is a proposition denoted by the VP or the small clause. In other words, exactly the same p-event is iterated (i.e., $q = p$). This is applicable to *tasi*.

The monosyllabic *tto* finds its restriction in the context: p and q provide answers to the contextually salient QUD.

How can we generalize and formalize this intuitive, loose notion of iteration across these two cases and what are its theoretical implications? Two thoughts:

- A loose definition that generalizes these as iteratives, just as we stated in the beginning of our talk. This account implies that context-sensitivity can parameterize the triggering mechanisms for a presupposition of the same type.
- A strict definition that separates the two lexical items. This account implies that context-sensitivity can identify presuppositions of different types that were once believed to be of the same type.

4.2 Non-canonical iterative readings of *tto*

Recall that only the monosyllabic *tto*, but not the bisyllabic *tasi*, is compatible with non-canonical iterative presuppositions such as the subjectless presupposition and the objectless presupposition. The felicity of *tto* is accounted for by its QUD-sensitivity.

(27) a. Subjectless presupposition with intransitive verb
QUD: Who died?
Phol-i **tto** cwuk-ess-ta.
Paul-NOM *tto* die-PST-DEC
'Paul died.' defined only if there is an earlier q-event addressing the QUD (e.g., 'John died')

b. Objectless presupposition with transitive verb
Context: Yesterday, Paul went to his favorite restaurant. He didn't eat anything all day beforehand. At the restaurant, he ate cherries first. Then, what did he eat?
Chikhin-ul **tto** mek-ess-ta.
chicken-ACC *tto* eat-PST-DEC
'He ate fried chicken.' defined only if there is an earlier q-event addressing the QUD (e.g., 'He ate cherries')

4.3 Unavailability of the restitutive reading of *tto*

The unavailability of the restitutive reading of *tto* is also predicted by its QUD-sensitivity. For instance, the sentence given in (28) is felicitous under the QUDs in (29).

(28) Mikeyl-un **tto** mwun-ul yel-ess-ta.
Miguel-TOP *tto* door-ACC open-PST-DEC
'Miguel opened the door again$_{\text{tto}}$.'

(29) Possible candidates for a contextually salient QUD for (28):

a. Who opened the door?
b. What did Miguel open?
c. What did Miguel do with the door?

The proposition corresponding to the repetitive reading —*Miguel had opened the door before*— successfully addresses one of the possible QUDs. However, the one corresponding to the intended restitutive reading —*The door had been open before*— fails to address any of them.

5 Conclusion

Empirical analyses:

- Both the bisyllabic iterative *tasi* and the monosyllabic *tto* are lexical presupposition triggers. The presuppositions triggered by these two items are sensitive to the structure to some extent.
- The presupposition triggered by the bisyllabic iterative *tasi* is entirely structure-sensitive.
- The presupposition triggered by the monosyllabic iterative *tto* requires a QUD in order to be completed. It is partially context-sensitive.

Theoretical implications:

- QUD-dependency/sensitivity can be a parameter not only for different types of presuppositions, but also for the same type of presuppositions.
- The distinction is not entirely dichotomic and there can be hybrid-type triggers occupying a middle ground, i.e., a lexically triggered presupposition that cannot be completed without a QUD.

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